Concept map - 8

Evolution of cloud computing connected hardware evolution. internet Software Evolution, distributed system and services delivery.

Evolution of cloud computing

- **mainframe computing (1980's - 1970's)**
  - centralized large computer.
  - multiple users via terminal
  - expensive
  - limited flexibility.
- **personal computing**
  - 1980's - 1990's
  - Individual PC's
  - local application
  - limited resource
  - Growth of desktop software
- **client server computing (1990's - 2000's)**
  - Dedicated server for multiple users
  - Better Resource sharing
  - Scalability improved
  - used in organizations
- **Distributed System 2000's**
  - multiple connected computers
  - share work and data
  - Improved reliability
  - foundation for cloud computing
- **virtualization 2000's**
  - one physical machine runs multiple virtual machine
  - Improved resource utilization
  - Enabled cloud environment
- **cloud computing (2010's - present)**
  - on - demand Resource
  - Scalable and elasticity
  - pay - as - you go
  - global access
  - Support modern applications.

DRIVER for Evolution:

- Need for higher performance
- Efficient Resource utilization
- growth of internet and web technology
- Increasing data and user demand
- advance in hardware & software

Enablers

- Hardware advance (multi-core cpu's, storage)
- Software advancy (virtualization)
- High Speed internet
- open standards and web services (APIs)
- Demand for Scalable and cost-efficient solution

Concept map-3

Servic models . map Iaas , pass, Sass, and xaas with provide Responsibility and users Responsibility

Cloud Service models

Iaas

Infrastructure as a Service

- Provide virtualized computing resource
- user can install and manage os - middleware and application
- High flexibility and control

Example

Amazon Ec2 microsoft Azure vms google compute engine

Provide responsibility

Physical infrastructure

User responsibility

- os, middleware application, data security configuration.

Paas

platform as a Service

- provide as a platform for application development and deployment
- Includes runtime environment middleware development tools
- user focus on writing and deploying code

Example

- google app engine microsoft Azure App Services

Provide Responsibility

Infrastructure os runtime environment middleware development tools.

Saas

Software as a Service

provide Ready to use software over the internet

- Not installation Required
- users acess via web Browser.
- provide manager everything.

Example

Gmail google Docs Scale force micro soft 365

Provide Responsibility.

Entire infrastructure platform Application and maintaine.

User Responsibility

User the application and manage their data.

Xaas

Anything as a Service

- general term for any Services over the cloud.
- Includes Daas, Paas, Saas, Naas etc
- provide specialized Services.

Examples.

Databases at a Service Function as a Service Storage as a Service

Provide Responsibility

Depends on the Specific Services

User Responsibility.

Depends on the Specific Services

4 Virtualization and web services at enabus of cloud show how those two concept supports cloud deployment Scalability

CONCEPT MAP - 4

Virtualization and webservices and enabus of cloud

cloud Deployment and Scalability

Virtualization

Key components

- virtual machines (vm)
- hypervisior (type 1 / type 2)
- Guest OS
- Resource allocation (cpu, ram)

Benefits

- Better Resource utilization
- Isolation and Security
- cost Reduction
- Flexibility and Scalability
- Easy deployment & management

Work Together

- virtualization provide the infrastructure
- web service provide that communication and access to those Resource.

Web service

- Standard ways for applications to communicate over a Network using Apis

Typy.

API (Applications programming Enterface)
Soap (protocol - based, xml)
REST (architectural style, uses. HTTP)
used (with format like xml, json

Benefits

Interoperability blw different platforms
Access to cloud functions and data.
Scalable and flexible integration.

cloud computing Fundamentals

cloud computing

is

Definition.

- A model for delivery computing resource (server storage, application, service over the internet
- provide on-demand access to shared Resource without owning physical infrastructure

Provide

+ Benifits
- Scalability, lower up front cost
- high availability & reliability
- Access from any where
- Automatic updates
- Better resource utilization
- Improved collabration
- Support Innovation

Delivery

Compute | Storage | Application | Services | via Internet

Examples:

- google Drive, AWS, microsoft Azure office 365, online Streaming
- cloud based online gaming platforms.

Key characteristics

- on demand self Service
- Resource pooling
- Board Network access
- Elasticity
- measured Service

Limitations

Internet dependency
vendor lockin
Possible downtime
compliance legal concern
Performance may vary based on Network